Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)



SOLUTIONS ARCHITECT

Enterprise Integration | Data Platforms | AI-Enabled Solutions

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PROFESSIONAL SUMMARY

Key results: **98%** fewer reported safety events (**50 to 1** per season) with a graph neural network system in production | **99.8%** availability across **200+ integration points and 86 systems** | data delivery from next-day to **near-real-time** for **78 consuming systems** | **EUR 8M** cumulative savings

Solution and integration architect and Data & AI lead with 20+ years in the enterprise IT of a large, transaction-intensive 24/7 airline group: designed, governed and delivered an integration backbone of **200+ integration points across 86 systems** (**~50M annual transactions**) on Kafka, IBM MQ and REST/SOAP APIs, from core operations to SAP finance, HR/payroll and DWH.

Data and AI end to end: an operational data store with canonical data model, master data management, ETL/ELT pipelines and data quality controls (~100K update records a day on average in peak season, 78 consuming systems across eleven downstream domains), and a graph neural network system in production for an **NP-hard** Quadratic Assignment Problem.

Sustained core operations through more than eightfold growth, from 10M to 85M passengers a year, as customer-side product owner for the core vendor operations suite, the B2B data exchange platform and planning platforms across three group companies: RFP/POC evaluations including a five-vendor RFP, technical due diligence and key-user training. A hands-on technical authority with **delivery accountability**; PMP-certified, rated **exceptional** in the DDI Manager Ready assessment.

WORK EXPERIENCE

Career Break and Professional Development | **Istanbul** | **Sep 2024 – Present**

- Planned break for family and structured upskilling: **Harvard's CS50 AI** (12 Python projects: search, optimization, machine learning, neural networks, NLP), AWS Certified AI Practitioner and Cloud Practitioner, PSPO I and continued research on graph-based AI for safety-critical operations.

Principal Solutions Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2023 – Sep 2024**

Official title: Distinguished Expert, the organization's highest technical grade.

- Cut reported safety events from look-alike and sound-alike identifiers by **98%**, from **50 to 1 per season** (sustained since), across up to ~1,500 daily peak-season operations, replacing 2010 heuristic assignments with an in-house graph-based similarity and assignment optimization system aligned with industry safety standards.
- Reframed identifier assignment as an **NP-hard** Quadratic Assignment Problem and built a scalable machine learning solution on **Graph Neural Networks (GNN)**: **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering over phonetic, visual and semantic similarity.
- Reduced monthly incidents from **8 to 1** in 10 months as owner of roadmaps and technical direction for **six mission-critical operational systems** (1,000 users, ~100 concurrent, 5M annual transactions, EUR 1M budget), sequencing change through **integration dependency management** and architectural constraints.
- Ran a core operations platform replacement selection (**5 vendors, 3 shortlisted, 2 POCs**; a **EUR 1.5M** decision) and vendor evaluation of three planning platforms through RFP/RFI, vendor management and business cases, advising VP-level leadership on build-versus-buy, technology strategy and long-term R&D strategy.
- Held data mismatches at **zero** across two companies and two core systems during a group company's migration from TROYA, the group's in-house reservations and inventory platform, to a separate third-party platform.
- Ran data synchronization, cross-company schedule distribution and change management in a transition architecture.
- Established design authority (architecture review board) and design reviews bringing decisions under ISO 27001 compliance, software quality management and enterprise architecture governance, supporting regulatory audits and compliance.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers) that delivered **9 systems to production**, as architecture and design decision owner, with Scrum Master support as required; rated team members' performance.
- Defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers.
- Shaped integration and technology strategy for next-generation operations platforms as cross-company subject matter expert for three group companies on the core scheduling and operations platform and on operations control systems.

Senior Integration Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

Official title: Senior Expert, Integration Architect.

- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model, with enterprise messaging and data streaming reference patterns adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Shortened new-integration lead time from **one week to one day**, cut integration-related incidents by **99%** and retired **47 legacy point-to-point interfaces** by reusing the group-wide reference patterns.
- Achieved **99.8% availability** (planned maintenance included) and **zero unplanned downtime** on mission-critical platforms serving **500+ concurrent users**, leading cross-functional delivery management across internal and external integration platforms through production deployment under high availability, disaster recovery design and SLA/SLO management.
- Eliminated dispatches into restricted zones (**18** in the 12 months before go-live, **zero** since) by engineering a real-time early warning system across two vendor platforms: **72** automated checks and alarms on plans for up to **~1,500 daily** peak-season operations.
- Enabled **1.5M** affected customers a year to rebook online, even those already en route, by extending core-platform integration with the in-house reservations platform TROYA to real-time exception and disruption management, propagating operations control decisions to reservation, inventory and customer-facing systems as they happened.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions**.
- Applied microservices, enterprise integration patterns, API-first design and API lifecycle management, including naming conventions, API versioning and backward compatibility.
- Engineered the reliability semantics behind the estate's **99.8%** availability: publish/subscribe and asynchronous processing, idempotency in consumers, retries and exponential backoff, timeouts and error recovery, dead-letter handling and poison-message processing, message ordering and durability, and replay of persisted streams for recovery.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Integrated a partner service behind an API gateway, handling API-key acquisition and token renewal on expiry.
- Raised release frequency from **1 to 4 a month** and **halved failed releases** by setting the estate's engineering practice: CI/CD architecture and DevOps release automation with Git-based (GitOps) configuration control, automated testing strategies, and performance and security testing gating releases.
- Ran **Scrum**-based agile delivery in **Jira** and **Confluence**: user stories with acceptance criteria, backlog management, sprint planning and Definition of Ready/Done.
- Caught **99%** of integration failures before users reported them and reduced mean time to detect from **one hour to one minute** with a real-time observability tool built across all live enterprise integrations.
- Based the tool on structured logging, correlation IDs and distributed tracing, and directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

Lead Solutions Architect and Platform Owner | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

Official title: Project Manager, Flight Operations.

- Designed and built a **Tier-0** (24/7 mission-critical) operational data store and evolved the operational plan data store (live since 2010) into a second Tier-0 platform.
- Took plan and operational-change delivery from **next-day to near-real-time** for **78 consuming systems**, replacing **70 separate data interfaces** by routing every change through both data stores, 24/7.
- Architected both as integration layers in front of two independent vendor platforms and as the authoritative **single source of truth** for enterprise reporting and downstream processing.
- Owned enterprise-wide operational data distribution for planning, operational and post-operation periods as the single integration point for **eleven downstream domains**: planning, workforce, load planning, reservation and customer-facing systems, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** on average in peak season, 24/7 and near-real-time, through source-to-target mappings, ETL/ELT pipelines and integration rules that consolidated, transformed and enriched every update across seasonal plans under data quality and metadata controls.
- Architected batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics.
- Set consumer-specific horizons, from a **rolling three-day window** to a **90-day window**.
- Fed operational changes (delays, cancellations) to customer-notification systems, triggering **75M notifications a year**, and operational data to loyalty and disruption hotel systems, so all three ran on the same data as operations.
- Designed operational data and integration architecture on the ODS pattern, standardizing core operational and plan data enterprise-wide: logical data models, entities and relationships, canonical data structures, ESB pattern applied twice.
- Evolved both Tier-0 platforms for **ten years** (2014 – Sep 2024) as de facto product owner with final authority over feature prioritization and architectural evolution.

- Coordinated operational stakeholders and cross-team dependencies for two Tier-0 platforms, carrying releases through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.
- Drove technical direction, integration and platform modernization for operational systems with a **EUR 6.5M** total programme budget (planning, scheduling, operations control) as lead architect.
- Worked end to end: system and business analysis, software architecture and hands-on coding of critical components through release, operational handover and 24/7 mission-critical production under ITIL.
- Managed the vendor platform's master and reference data with the team: built and maintained the tables where users edit it and trained key users in master data management (MDM), keeping master data owned at its source.
- Built a reporting layer into both platforms on the same enriched data and fed it back into the source vendor platforms, so enterprise reporting and operations shared one version of the data.
- Kept **850 users** working on the core operations platform throughout the pandemic by coordinating, as its product owner, the infrastructure teams' VMware-based solution.

Integration Solution Design Lead | **Turkish Airlines** | **Istanbul, Turkiye** | **Jan 2007 – Jun 2014**

Official title: Acting Team Leader.

- Delivered **EUR 8M** in cumulative savings from 2012 to 2023 (**EUR 800K** in the first year) with a multi-currency cost reconciliation engine built on a data pipeline of entry/exit timestamps and segment distances.
- Led a **6-person team** through the lifecycle of **100+ integrations** (requirements and system analysis, design, development, testing, documentation), harmonizing project plans and tracking cross-functional milestones per project.
- Rated team members' performance and interviewed technical candidates as Acting Team Leader, with hiring decisions made by others.
- Architected the message integration and data mapping from core planning data to the in-house reservations platform (TROYA) under industry messaging standards, keeping inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated services and reason codes.
- Introduced (2007) and operated a standards-based B2B data exchange platform for recurring data interchange with **54 partners**.
- Headed solution design and delivery for operations, workforce and planning integrations across global technology vendors, building multithreaded real-time integration frameworks (2010) and contributing to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Built a plan data store in 2010 to cut mainframe dependency: plans from a new vendor planning system landed in Oracle and the data warehouse, easing SAP integration; the efficiency team rated them far more accurate than the earlier vendor feed.
- Developed the first heuristic similarity and assignment system (phonetic, visual and semantic similarity), delivered in 2010: the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by the head of operations for operations projects, the identifier-safety project first among them, and personally by the **General Manager and CEO**.

Software and Integration Engineer | **Turkish Airlines** | **Istanbul, Turkiye** | **Apr 2000 – Jan 2007**

Official title: Analyst Programmer.

- Contributed to the first production implementation of a distributed vendor operations suite, later owned for two decades.
- Cut costs by **EUR 100K** a year by designing and building the timetable publication system from scratch: worldwide schedules and partner flights in, a PDF timetable for web download and print out, so the scheduling department began preparing the timetable books itself (Java, C, PL/SQL).
- Administered the suite at application level with 24/7 availability responsibility (upgrades and patches on AIX, database schema upgrades) and built its bi-directional real-time interfaces to the IBM mainframe on IBM MQ.
- Drove the technical work of the first major legacy modernization from day one as an implementation team member, moving scheduling and operations control off the IBM mainframe (ADABAS/Natural) to Oracle on Unix (AIX) and the vendor suite.
- Designed and built a 24/7 real-time world map tracing every company flight in the air with simulated positions: architecture decisions, mainframe integration over IBM MQ and UDP multicast (Java, Oracle, ILOG JViews Maps).
- Ran **statistical analysis** of historical data to determine the methodology for the project that calculated average passenger and baggage weights; the methodology was adopted for weight-and-balance calculations.

TECHNICAL EXPERTISE (hands-on) & CORE COMPETENCIES

Integration Architecture and Schema Governance: Event-Driven Architecture, API Architecture, reference architectures, Apache Kafka with schema registry, schema versioning and backward-compatibility rules, controlled change across event-driven integrations, IBM MQ, Enterprise Service Bus (ESB), REST/SOAP APIs; GraphQL (design level).

Data Architecture and Governance: Enterprise Data Architecture, Data Modeling, Canonical Data Model, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture, master data management (MDM), data lineage through source-to-target mappings, data quality and metadata controls.

Data and AI Platform Architecture: Graph data models (Neo4j), GNN-based assignment and decision support (PyTorch, PyTorch Geometric), data pipelines, MLOps, data warehouse and BI feeds, real-time analytics.

Engineering Platforms & Languages: Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML; multithreaded real-time and database-driven systems; 50+ applications and integrations to DWH, SAP and MRP, with incident analysis and performance tuning.

Executive Leadership and Delivery: Digital transformation, transformation governance, executive (C-level) stakeholder management, PMI/PMP delivery governance, operational readiness, customer consulting and workshop facilitation.

EDUCATION

• Ph.D. (coursework)	Business Administration (Finance)	Yeditepe University	Istanbul
• M.Sc.	Management Engineering; thesis on data warehousing for mass customization	Istanbul Technical University	Istanbul
• B.Sc.	Computer Science Engineering	Istanbul University	Istanbul

LANGUAGES

• English	C1	Full Professional
• German	A2	Studying towards B1
• Turkish		Native

CERTIFICATIONS

• Project Management Professional (PMP), Credential ID 1225730	PMI	
• Professional Scrum Product Owner I (PSPO I), Credential ID 1347372	Scrum.org	2026
• ITIL v3 Foundation, Candidate No. 990453483	BCS, The Chartered Institute for IT	
• AWS Certified Cloud Practitioner (CLF-C02)	Amazon Web Services (AWS)	2026
• AWS Certified AI Practitioner (AIF-C01)	Amazon Web Services (AWS)	2026
• CS50's Introduction to Artificial Intelligence with Python (12 projects)	Harvard University (CS50)	2026
• Manager Ready (DDI) leadership simulation: exceptional rating	TAG Taurus Group	2024
• AI, Cloud and Container Foundations: OCI AI Foundations; Agentic AI Foundations; IBM SkillsBuild; PMI AI (CPMAI, GenAI); Docker Foundations Professional Certificate	Oracle, IBM, PMI, Docker	2025
• Lufthansa Systems SchedConnect vendor training (two weeks)	Lufthansa Systems	